

1. Embedded hardware/software systems are basically designed to
 - (a) Regulate a physical variable
 - (b) Change the state of some devices
 - (c) Measure/Read the state of a variable/device
 - (d) Any/All of these
2. Little Endian processors
 - (a) Store the lower-order byte of the data at the lowest address and the higher-order byte of the data at the highest address of memory
 - (b) Store the higher-order byte of the data at the lowest address and the lower-order byte of the data at the highest address of memory
 - (c) Store both higher order and lower order byte of the data at the same address of memory
 - (d) None of these
3. An integer variable with value 255 is stored in memory location at 0x8000. The processor word length is 8 bits and the processor is a big endian processor. The size of integer is considered as 4 bytes in the system. What is the value held by the memory location 0x8000?
 - (a) 0xFF
 - (b) 0x00
 - (c) 0x01
 - (d) None of these
4. The instruction set of RISC processor is
 - (a) Simple and lesser in number
 - (b) Complex and lesser in number
 - (c) Simple and larger in number
 - (d) Complex and larger in number

5. Which of the following is true about CISC processors?
- (a) The instruction set is non-orthogonal
 - (b) The number of general purpose registers is limited
 - (c) Instructions are like macros in C language
 - (d) Variable length Instructions
 - (e) All of these
 - (f) None of these

6. Which of the following processor architecture supports easier instruction pipelining?
- (a) Harvard (b) Von Neumann (c) Both of them (d) None of these
7. Microprocessors/controllers based on the Harvard architecture will have separate data bus and instruction bus. This allows the data transfer and program fetching to occur simultaneously on both buses. State True or False
- (a) True (b) False
8. Which of the following is one-time programmable memory?
- (a) SRAM (b) PROM (c) FLASH (d) NVRAM
9. Which of the following memory type is best suited for development purpose?
- (a) EEPROM (b) FLASH (c) UVEPROM
- (d) OTP (e) (a) or (b)

10. EEPROM memory is alterable at byte level. State True or False
(a) True (b) False
11. Non-volatile RAM is a Random Access Memory with battery backup. State True or False
(a) True (b) False
12. Execution of program from ROM is faster than the execution from RAM. State True or False
(a) True (b) False
13. Dynamic RAM stores data in the form of voltage. State True or False
(a) True (b) False

14. The control algorithm (Program instructions) and or the configuration settings that are kept in the code (Program) memory of the embedded system are known as Embedded Software. State True or False
(a) True (b) False
15. Which of the following is an example for Wireless Communication interface?
(a) RS-232C (b) Wi-Fi (c) Bluetooth
(d) IEEE1394 (e) both (b) and (c)
16. Which of the following is (are) examples for Application Specific Instruction set Processor(s)
(a) Intel Centrino (b) Atmel Automotive AVR
(c) AMD Turion (d) Microchip CAN PIC (e) All of these (f) both (b) and (d)
17. How many memory cells are present in 1Kb RAM
(a) 1024 (b) 8192 (c) 512 (d) 4096
(e) None of these

Module III

Characteristics & Quality Attributes of Embedded Systems

Characteristics of Embedded System

Each Embedded System possess a set of characteristics which are unique to it. Some important characteristics of embedded systems are:

- ☐ Application & Domain Specific
- ☐ Reactive & Real Time
- ☐ Operates in 'harsh' environment
- ☐ Distributed
- ☐ Small size and Weight
- ☐ Power Concerns

Application & Domain Specific

- Embedded Systems are developed to do a specific task.
- An embedded control unit in a washing machine will not work in microwave
- An embedded control unit developed for a particular domain cannot replace an ECU with another domain.

Reactive & Real Time

- ES are reactive in nature as they produce change in output for change in input.
- ES interact with real world through input sensors. Any change in real world often called event triggers the ES to run a control algorithm to regulate the output.
- The event can be periodic, occurs at regular intervals of time or aperiodic (unpredicted) can occur any time.
- For periodic events on chip timer or RTC can be used to monitor the inputs.
- Unpredicted inputs are monitored through interrupts.

- Real Time ES means the response to change in input is time bound.
- Based on time critically ES can be
 - a. Soft Real Time System (RTS) : Here response time is flexible. Failing to maintain time does not cause any major hazard.

Eg: Bank ATM

- b. Hard Real Time systems: The response time is very critical here. If ES takes longer time to react the result can be very catastrophic.

Eg: Antilock Braking System

Operates in 'harsh' environment

- ES deployed in unfavourable/harsh environment should be robust and withstand the adverse operating conditions.

An ES in oil well, coal mines or nuclear power plant

- The developers should design an ES based on where it will be deployed.
- ES should be able to withstand variation in temperature, absorb shock , withstand power fluctuations, corrosion.

Distributed

- Distributed means that ES may be part of larger Embedded Systems. Many distributed ES may form one large ES
- Eg: ATM
 - 1. Card Reader: Reads the card and validates
 - 2. Transaction unit: To perform the transaction
 - 3. Currency Counter: To count the currency
 - 4. Printer: To print transaction details

All these can be visualised as independent ES which work together.

Small Size and Weight

- Product aesthetics are very important while choosing a product.
- A mobile is expected to be sleek, light weight and stylish too.

Power Concerns

Power is another important concern

Dissipate less heat, for more heat producing ES cooling system is required.

In battery operated devices low power components are used.

Characteristics & Quality Attributes of Embedded Systems

Quality Attributes of Embedded Systems:

Represent the non-functional requirements that needs to be addressed in the design of an embedded system. The various quality attributes that needs to be addressed in any embedded system development are broadly classified into

☐ Operational Quality Attributes

Refers to the relevant quality attributes related to embedded system when it is in the operational mode or '**online**' mode

☐ Non-Operational Quality Attributes

The Quality attributes that needs to be addressed for the product 'not' on the basis of operational aspects are grouped under this category

Characteristics & Quality Attributes of Embedded Systems

Operational Quality Attributes

- ☐ Response
- ☐ Throughput
- ☐ Reliability (MTBF & MTTR)
- ☐ Maintainability
- ☐ Security
- ☐ Safety

Response

- It is measured as the quickness of the system.
- Based on the response time the ES are classified as soft and hard real time systems.

Throughput

- It deals with efficiency of a system. It can be defined as the rate of production or operation of a defined process over a period of time. Throughput is measured in terms of Benchmark. A Benchmark is reference point by which something can be measured. It can be a set of performance criteria.

Reliability

- It's a measure of how much % you can rely upon the proper functioning of the system
- MTBF Mean Time Between Failures
- MTTR Mean Time To Repair

Are used to define the reliability of the system

MTBF defines the frequency of failures in hours/week/months

MTTR allows how long the system is allowed to be out of order following a failure. For an ES with critical application it should be order of minutes.

Maintainability

- Deals with support and maintenance provided to the owner of ES.
- Reliability and maintainability are complementary if an ES is reliable then needs less maintenance and vice versa.
- Maintenance can be

1. Scheduled or periodic Maintenance

ES may be made of consumable component which have to be regularly replaced. The period depends on usage or output produced.

Eg: Ink in printer

2. Maintenance to unexpected failures

Availability of ES in the ideal condition is

$$A_i = \text{MTBF} / (\text{MTBF} + \text{MTTR})$$

Security

- Confidentiality, Integrity and Availability are key issues
- Confidentiality : Protection of data and application from unauthorised disclosure
- Integrity : Protection of data and application from unauthorised modification
- Availability : Protection of data and application from unauthorised users
- Eg: PDA

Safety

- How safe is the system to user? Public? Environment?
- Eg: EMI radiation from mobile

Characteristics & Quality Attributes of Embedded Systems

Non-Operational Quality Attributes

- ☐ Testability & Debug-ability
- ☐ Evolvability
- ☐ Portability
- ☐ Time to Prototype and Market
- ☐ Per Unit and Total Cost

- **Testability:** How easily the ES can be tested
The ease with which the hardware and software can be tested
- **Debug-ability:** An ES system should have debug ability feature i.e hardware debug can be provided by some ports through which the hardware operation can be provided.
Software debugging can be done by WDT, Reset features.

- **Evolvability:**

An ES should be modifiable, upgradable w.r.t both Hardware and software.

Eg: New releases are often improved versions of the previous release

- **Portability:**
- The ease with which an ES can be ported to a different platform.
- Ease with which it can work the same with change in core/Firmware/ OS
- The firmware is portable if coding is done in embedded C and difficult if developed in assembly.
- Some applications are OS dependent and fail to work if the Os is changed

- **Time to market:**

The time elapsed between the time when the idea was conceptualised to when the market is ready for release.

Owing to excessive competition , continuous change in some ES (mobile) the time to market will be very narrow(days-months).

Time to prototype is also very important, if the prototype is developed faster then testing , bulk production can be done faster.

Time to prototype can be minimised by “Design Reuse” method

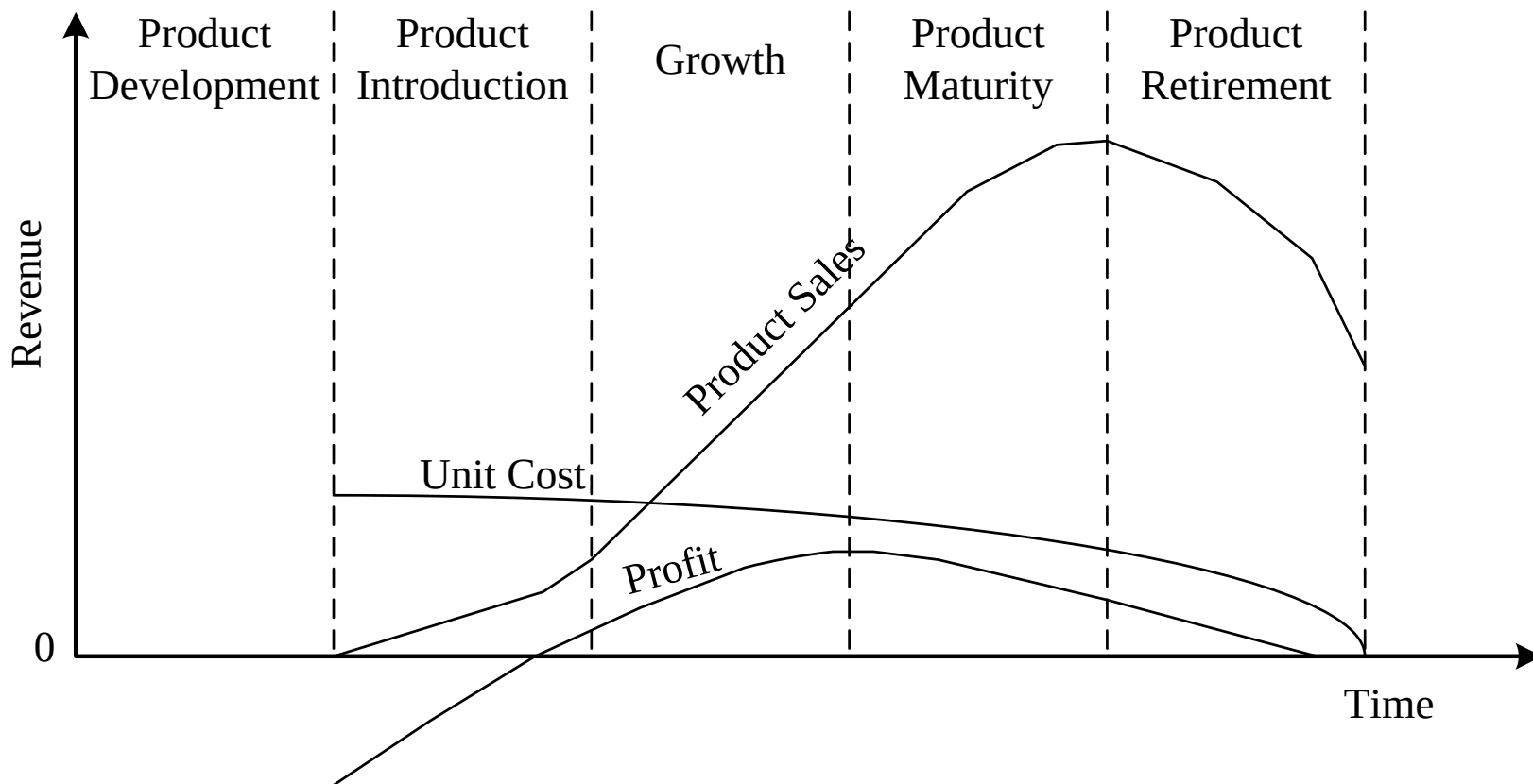
Per Unit Cost and Revenue

- Cost affects both developer and customer.
- Proper market study and cost benefit analysis should be carried out before fixing the price of the ES
- Every product has a product life cycle which involves design and development phase.
- The product idea generation, prototyping, roadmap definition, actual product design and development are different activities during this phase.
- During design and development there is only investment and no revenue
- Once the product is ready it is introduced to the market and this is called product introduction stage where sales and revenue is minimum

- During the growth the product tries to catch up with market and sales and revenue improve
- During product maturity the growth is steady and the revenue is at peak
- During product retirement the growth and market share falls and revenue starts dipping, at some point the manufacturer withdraws the product from the market.
- So its very important to hit the market earlier so that product can be active for a longer time in market.
- Reasons for lesser growth, lesser market share and lesser profit may faulty product, more competitors , high pricing etc.

Characteristics & Quality Attributes of Embedded Systems

The Product Life Cycle (PLC) Curve



What is Cost per Unit (C) ?

- NRE : Non Recurring Engineering Cost (NE)
- N: No of units developed
- Unit cost (U): Cost to develop one unit

$$C = U + \text{NRE}/N$$

If for a product, NRE =Rs10000, N=100, U=100

Then $C = 100 + 10000/100 = \text{Rs } 200$ per unit

If N=1000

$$C = 100 + 10000/1000 = \text{Rs } 110 \text{ per unit}$$

If for a product, NRE =Rs1000, N=100, U=100

Then $C = 100 + 1000/100 = \text{Rs } 110$ per unit

If N=1000

